After a certain period, the difference between the amount and the money deposited is called the compound interest which is abbreviated as C.I. Clearly, the compound interest will be more than the simple interest just because the principal keeps on changing for every time period.

We call the time period after which the interest is added to the principal, as the conversion period. For example, if the interest is compounded say quarterly, there will be four conversion periods in a year, every 3 months. In such cases, the interest rate will be one fourth of the annual rate and the number of times that interest will be compounded is four times the number of years.

In case of simple interest, the principal remains the same for the whole duration where as in case of compound interest, the principal keeps on changing as per the conversion period. The simple interest and the compound interest remains the same for the first conversion period.

### Illustration 1

To find the compound interest on ₹20000 for 4 years at 10 % p.a compounded annually and compare it with the simple interest obtained for the same.

#### Calculating Compound Interest

Principal for the I year = ₹20000

Interest for the I year $\left(\frac{20000\times10\times1}{100}\right)$ = ₹ 2000

Amount at the end of I Year (P+I) = ₹22000

That is, Principal for the II year = ₹22000

Interest for the II year $\left(\frac{22000\times10\times1}{100}\right)$ = ₹ 2200

Amount at the end of II year = ₹24200

That is, Principal for the III year = ₹24200

Interest for the III year $\left(\frac{24200\times10\times1}{100}\right)$ = ₹ 2420

Amount at the end of III year = ₹26620

That is, Principal for the IV year = ₹26620

Interest for the IV year $\left(\frac{26620\times10\times1}{100}\right)$ = ₹ 2662

∴ Amount at the end of IV year = ₹29282

∴ Compound Interest for 4 years = A – P = 29282 – 20000 = ₹9282

#### Calculating Simple Interest

Recall that,

Simple Interest, $I = \frac{PNR}{100}$

Here, P = ₹20000

N = 4 years

R = 10%

$\therefore I = \frac{20000\times4\times10}{100}$

That is, I = ₹ 8000

What we observe from the calculation of C.I is a repeated multiplication by the factor 1.1 as 20000 (×1.1), 22000 (×1.1), 24200 (×1.1), 26620 (×1.1) for 4 years. We also note that the compound interest (₹9282) grows faster and is clearly more than the simple interest (₹8000) obtained. When the time period is longer, the above method is time



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